

Annotation Guide

This annotation guide was developed to support the consistent and reproducible annotation of pavement deformations in the PDD2026 dataset. The guide provides standardized annotation instructions for the five pavement deformation categories included in the dataset: **Depression, Rutting, Upheaval, Pothole, and Patching**. The deformation definitions were derived from ASTM D6433 pavement distress terminology and subsequently reviewed by pavement engineering experts during the dataset development process. The primary objective of this guide is to ensure that annotators apply consistent interpretations when identifying pavement deformations and delineating deformation boundaries at the pixel level.

For each deformation category, the guide provides:

1. A concise deformation definition.
2. Visual characteristics commonly observed in field imagery.
3. Criteria describing conditions that should be annotated.
4. Boundary annotation rules for determining polygon limits.

Annotations were performed using polygon-based pixel-level segmentation. Annotators were instructed to delineate only the visible extent of each deformation and to avoid estimating boundaries obscured by vehicles, shadows, vegetation, pedestrians, or other objects. Polygon boundaries were determined according to the visible physical characteristics of the deformation rather than surrounding pavement texture or color variations.

The annotation process was designed to support deformation-specific segmentation and detailed pavement condition assessment. Therefore, when multiple deformation types were visually distinguishable within the same image, each deformation was annotated according to its respective class definition. This approach preserves detailed pavement condition information and enables the identification of different deformation mechanisms that may occur within the same road section.

Because real-world pavement surfaces often exhibit complex deterioration patterns, additional annotation rules are provided for overlapping deformations, adjacent deformations, and other ambiguous boundary situations. These rules were established to improve annotation consistency, support reproducibility, and facilitate future dataset reuse or reannotation efforts.

The purpose of this guide is to provide a transparent annotation protocol that supports reproducible computer vision research, benchmarking of pavement deformation segmentation methods, automated road inspection, pavement condition assessment, infrastructure monitoring, and the development of advanced pavement deformation analysis techniques.

A. Pavement Deformation Classes

This section describes the five pavement deformation categories included in the PDD2026 dataset: Depression, Rutting, Upheaval, Pothole, and Patching. The class definitions were derived from ASTM pavement distress terminology and refined through expert review to ensure consistent interpretation during the annotation process.

For each deformation category, this guide provides a definition, visual characteristics, inclusion criteria, exclusion criteria, boundary annotation rules, and representative examples. These guidelines were established to improve annotation consistency, reduce inter-annotator variability, and support reproducible pavement deformation segmentation research.

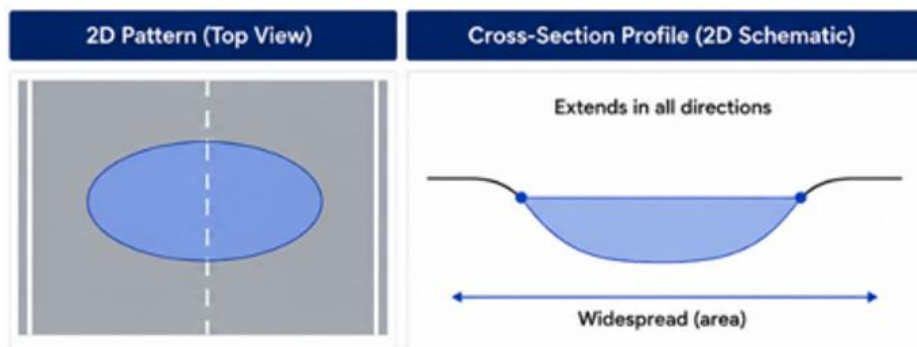
The objective of the class-specific annotation rules is to ensure that visually distinguishable deformation mechanisms are consistently identified and delineated at the pixel level. When multiple deformation types coexist within the same road section, each deformation should be evaluated according to its respective class definition and annotation criteria.

1. Depression

Depression commonly appears as a localized sunken area with visible surface settlement. The deformation boundary may be gradual rather than sharply defined. In many cases, the depressed area is accompanied by secondary distress features such as surface cracking caused by differential settlement. Depending on the severity of deformation, slight uplifting or heaving may also be observed around the perimeter of the depressed region due to displacement of surrounding pavement materials.

Visual Characteristics

Depression commonly appears as a localized sunken pavement area caused by surface settlement. The affected region may exhibit gradual boundary transitions and often forms an irregular shape. Depending on the severity of deformation, water ponding, cracking, or minor uplift around the perimeter may be observed. Unlike rutting, depression does not necessarily follow the direction of traffic and may extend in multiple directions across the pavement surface.



Criteria:

- ✓ Clearly visible localized surface settlement
- ✓ Downward displacement relative to the surrounding pavement
- ✓ Water ponding within a shallow basin
- ✓ Depressed area with visible cracking associated with the deformation
- ✓ Depressed area accompanied by minor uplift around the boundary
- ✓ Surface remains largely intact without substantial material loss

Boundary Annotation Rule

Annotate the entire visible depressed area by following the outer extent of the surface settlement. Include cracked regions, ponded water, and surrounding deformation features that are visually associated with the same depression. Minor uplift occurring at the edge of the depression may be included when it forms part of the same deformation mechanism.

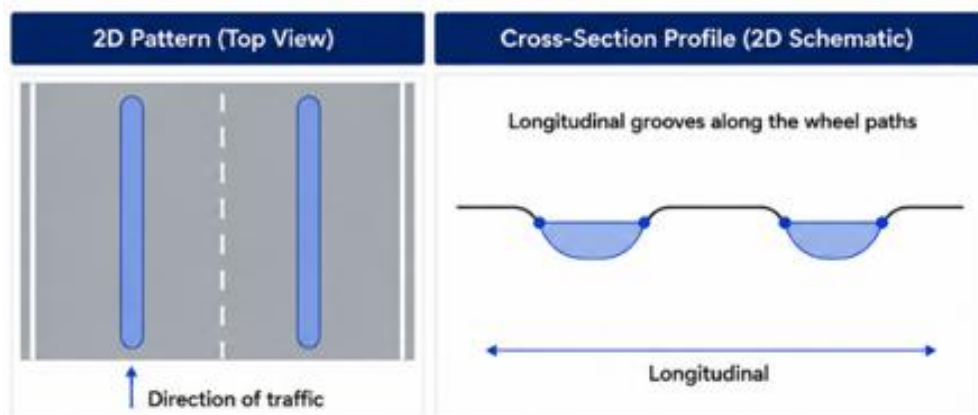
If a visually distinguishable pothole exists within the depression area, both deformities should be annotated independently according to their corresponding class definition.

2. Rutting

Rutting is a longitudinal pavement deformation characterized by permanent surface depressions that develop along the wheel paths of repeated traffic loading. Rutting typically appears as elongated grooves parallel to the direction of travel and results from the accumulation of permanent deformation within the pavement structure or underlying layers.

Visual Characteristics

Rutting commonly appears as one or more continuous longitudinal depressions aligned with traffic flow. The deformation is typically confined to wheel paths and may extend over a considerable distance. In severe cases, rutting may be accompanied by water accumulation within the grooves after rainfall. Minor cracking, surface wear, or edge deformation may also occur adjacent to the rutted area. Unlike depression, rutting exhibits a distinct linear pattern that follows the vehicle travel direction.



Criteria:

- ✓ Clearly visible longitudinal grooves along wheel paths
- ✓ Surface deformation aligned with the traffic direction
- ✓ Continuous or semi-continuous wheel-track depressions
- ✓ Parallel or single depressions corresponding to left and/or right wheel paths
- ✓ Rutting accompanied by water accumulation within the wheel paths
- ✓ Rutting associated with minor cracking or edge deformation

Boundary Annotation Rule

Annotate the entire visible extent of the rutting area by following the boundaries of the longitudinal wheel-path deformation. The polygon should encompass the full length and width of the pavement surface exhibiting permanent downward deformation that can be visually distinguished from the surrounding pavement.

If two wheel-path ruts are clearly separated, each rut may be annotated as an independent instance. Conversely, if the deformation forms a continuous rutting region connecting both wheel paths, a single annotation may be used to represent the entire affected area.

In some cases, rutting may be accompanied by localized or elongated raised pavement surfaces caused by material displacement along the sides of the wheel path. The areas exhibiting downward deformation shall be annotated as Rutting, whereas the areas exhibiting upward pavement displacement shall be annotated as Upheaval. These deformation types shall be separated according to the visible transition between the depressed and elevated pavement surfaces and should not be merged into a single annotation.

When potholes, patching, depressions, or other pavement deformation types occur within or adjacent to a rutted region and remain visually distinguishable, each deformation shall be annotated independently according to its corresponding class definition. Overlapping annotations are permitted when multiple deformation mechanisms coexist within the same pavement area and can be clearly differentiated based on their visual characteristics.

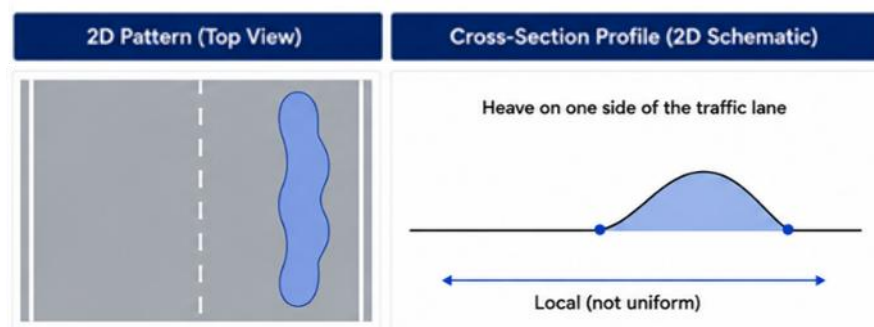
3. Upheaval

Upheaval is a localized upward displacement of the pavement surface caused by material expansion, displacement, or underlying structural instability. The deformation appears as a raised region that protrudes above the surrounding pavement surface and may occur independently or in conjunction with other pavement deformation types.

Visual Characteristics

Upheaval is commonly characterized by:

- A visibly elevated pavement surface relative to the surrounding roadway.
- Localized bumps, humps, or raised ridges.
- Abrupt or gradual changes in pavement elevation.
- Surface distortion that alters the normal pavement profile.
- Raised areas adjacent to other pavement deformations, such as rutting or depression.
- Shadows, highlights, or texture changes that indicate upward surface displacement.



Criteria

- ✓ Clearly visible upward displacement of the pavement surface.
- ✓ Localized bumps, humps, or raised pavement regions.
- ✓ Longitudinal or irregular raised areas resulting from material displacement.
- ✓ Pavement sections exhibiting elevation higher than adjacent pavement surfaces.
- ✓ Raised ridges formed adjacent to rutting caused by lateral movement of pavement materials.
- ✓ Upheaval occurring independently or in proximity to other pavement deformation types.

Boundary Annotation Rule

Annotate the entire visible extent of the raised pavement region by following the boundaries of the upward deformation. The polygon should encompass all pavement areas exhibiting elevation above the surrounding surface that can be visually distinguished from adjacent pavement.

The annotation boundary should be determined using the transition between the elevated pavement and the surrounding normal or deformed pavement surfaces. Local texture variations alone should not be used to define annotation boundaries. In some cases, upheaval may occur adjacent to rutting or depression. The uplifted area shall be annotated as Upheaval, while neighbouring pavement regions exhibiting downward deformation shall be annotated according to their respective classes. The boundaries between adjacent deformation types should be established based on the visible transition between elevated and depressed pavement surfaces.

When potholes, patches, depressions, rutting, or other deformation types occur within or adjacent to an upheaval region and remain visually distinguishable, each deformation shall be annotated independently according to its corresponding class definition. Overlapping annotations are permitted when multiple deformation mechanisms coexist within the same pavement area and can be visually differentiated.

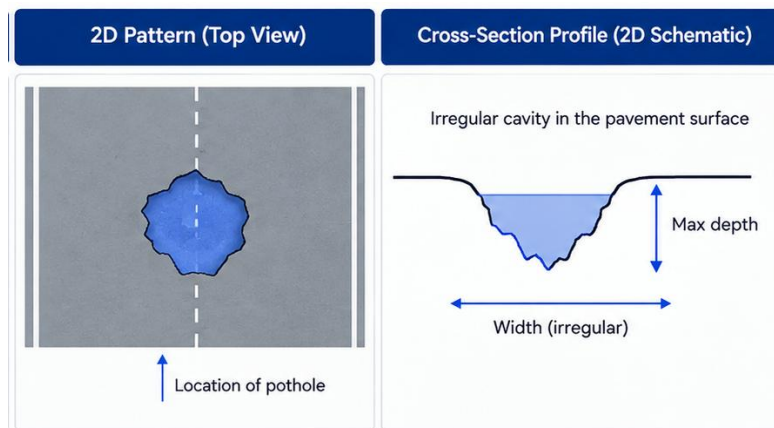
4. Pothole

A Pothole is a localized pavement defect characterized by the loss of pavement material, resulting in a distinct cavity or hole on the pavement surface. Potholes may develop independently or occur within other pavement deformations, such as depressions, rutting, or patched areas. Unlike depression or rutting, potholes are specifically associated with visible material loss and cavity formation.

Visual Characteristics

Potholes are commonly characterized by:

- Visible loss of pavement material.
- Distinct cavities with recognizable boundaries.
- Irregular shapes and varying sizes.
- Exposed aggregate, base material, or underlying pavement layers.
- Dark shadows or water accumulation within the cavity.
- Abrupt surface discontinuities relative to the surrounding pavement.
- Occurrence within larger deformation areas such as depression or patching.



Criteria

- ✓ Clearly visible pavement cavities resulting from material loss.
- ✓ Bowl-shaped or irregular holes with identifiable boundaries.
- ✓ Localized defects exposing aggregate, base material, or underlying pavement layers.
- ✓ Cavities containing standing water when the pothole boundary remains visible.
- ✓ Potholes occurring within depression, rutting, or patching regions.
- ✓ Material-loss cavities that remain visually distinguishable from surrounding pavement deformations.

Boundary Annotation Rule

Annotate the entire visible extent of the material-loss cavity by following the observable pothole boundary. The polygon should encompass only the area affected by pavement material loss and should not include surrounding pavement surfaces that remain structurally intact.

The annotation boundary shall be determined based on the visible edge of the cavity rather than surrounding shadows, water stains, debris, or texture variations. Water-filled potholes should be annotated according to the visible cavity boundary when the pothole limits can still be identified.

If multiple potholes are separated by intact pavement surfaces, each pothole shall be annotated as an independent instance. Conversely, interconnected cavities forming a continuous material-loss region may be annotated as a single pothole instance.

When portions of the pothole boundary are obscured by vehicles, shadows, vegetation, pedestrians, or other objects, only the visible cavity area shall be annotated, and hidden boundaries should not be estimated.

When potholes occur within a larger depression, rutting, or patched region and remain visually distinguishable, each deformation shall be annotated independently according to its corresponding class definition. Overlapping annotations are permitted when multiple pavement deformation mechanisms coexist within the same pavement area and can be visually differentiated.

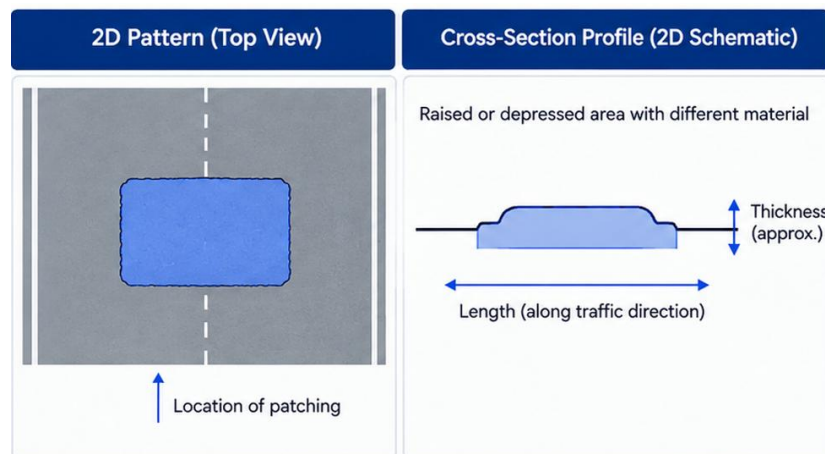
5. Patching

Patching is a repaired pavement area where defective pavement material has been removed, covered, or replaced with repair material to restore the road surface. The repaired area is typically distinguishable from the surrounding pavement by differences in texture, color, material composition, surface roughness, or repair boundaries.

Visual Characteristics

Patching is commonly characterized by:

- A repaired pavement surface with visible differences in color, texture, or material properties.
- Clearly identifiable repair boundaries.
- Rectangular, polygonal, linear, or irregular repair shapes.
- Surface appearance that differs from the surrounding original pavement.
- Areas indicating previous maintenance or pavement rehabilitation activities.
- Repaired regions that may contain subsequent pavement deterioration.



Criteria:

- ✓ Clearly visible repaired pavement sections.
- ✓ Areas exhibiting distinct repair boundaries relative to the surrounding pavement.
- ✓ Pavement surfaces with noticeable differences in texture, color, aggregate composition, or surface finish resulting from maintenance activities.
- ✓ Localized or extensive repair regions created to restore damaged pavement.
- ✓ Patched areas that remain visually distinguishable despite aging or weathering.
- ✓ Repaired pavement sections containing other deformation types, provided that the repair boundary remains visually identifiable.

Boundary Annotation Rule

Annotate the entire visible repaired area by following the repair boundary that separates the patched pavement from the surrounding original pavement. The polygon should encompass only the pavement surface that clearly belongs to the patching region.

Annotation boundaries should be determined primarily based on visible repair limits, including differences in texture, material composition, surface finish, or construction joints. Color differences alone should not be used as the sole criterion for defining patch boundaries. When multiple patches are located adjacent to one another, each patch shall be annotated as an independent instance if a visible repair boundary can still be identified between them. Adjacent patches should not be merged into a single annotation solely because they are physically connected or located in close proximity.

A single patching annotation may only be used when the repaired area forms a continuous maintenance region and no distinguishable boundary exists between neighboring patches. If portions of a patch boundary are obscured by vehicles, shadows, vegetation, pedestrians, or other objects, only the visible repair area shall be annotated, and hidden boundaries should not be estimated.

B. Annotation Rules for Complex Pavement Conditions

In real-world pavement conditions, multiple deformation types may occur within the same road section. These situations can be categorized into two cases: overlapping distresses and adjacent distresses. To ensure annotation consistency and reproducibility, specific labeling rules were established for both scenarios.

1. Overlapping Distresses

Overlapping distresses occur when one pavement deformation is physically contained within another deformation and both distress mechanisms remain visually distinguishable. In such cases, each distress should be annotated independently according to its own visible boundary.

A typical example is a **pothole located inside a larger depressed area**. The pothole represents localized material loss, whereas the depression represents a broader surface settlement without complete material removal. Since these are different physical distress mechanisms, both should be annotated separately. The depression boundary should encompass the entire depressed region, while the pothole boundary should delineate only the cavity associated with material loss.

This annotation strategy preserves the full structural information of pavement deterioration and enables more detailed distress analysis.

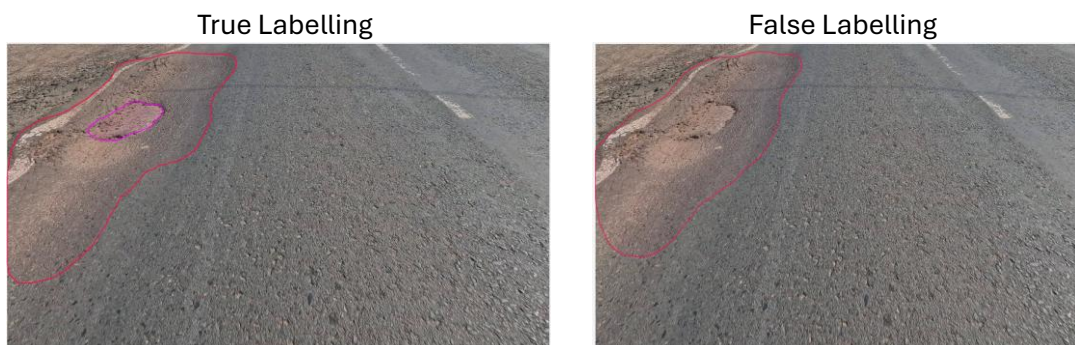


Figure 1. Example of Overlapping Distresses (Depression-Red Line and Pothole-Purple Line)

True Labelling: The depression is annotated over the entire visible settlement area, while the pothole is annotated separately within the depression according to the visible material-loss boundary. Both distresses are preserved as independent instances.

False Labelling: Only the pothole is annotated while the surrounding depression is ignored, or both distresses are merged into a single annotation. These approaches fail to represent the true pavement condition and reduce the structural information available for analysis.

2. Adjacent Distresses

Adjacent distresses occur when two or more pavement deformations are located next to each other while remaining physically separate and visually distinguishable. In such situations, each distress shall be annotated independently by following its respective visible boundary. Distresses that are adjacent should not be merged into a single annotation simply because they are in close proximity.

A typical example is a depression located adjacent to an upheaval, as shown in Fig. 2. Although both distresses occur within the same road section, they represent different pavement deformation mechanisms. The depression corresponds to a localized surface settlement, whereas the upheaval corresponds to an upward displacement of the pavement surface. Therefore, the depression boundary should encompass only the visibly depressed area, while the upheaval boundary should follow the limits of the raised pavement region.

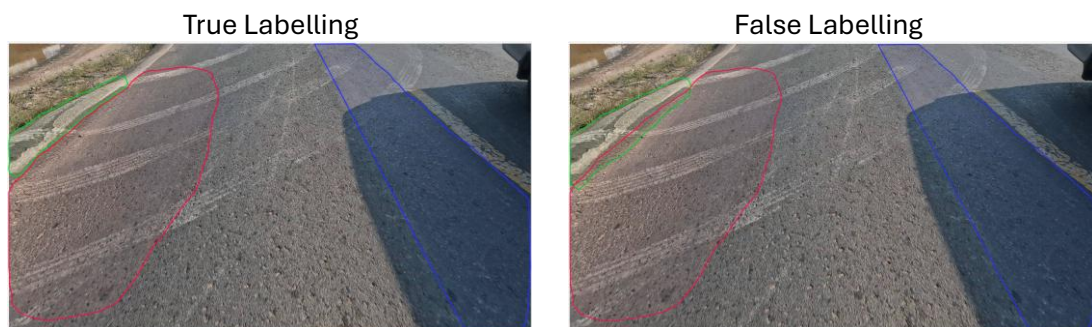


Figure 2. Example of Adjacent Distresses (Depression-Red line and Upheaval-Green Line)

This annotation strategy preserves the physical characteristics of each deformation type, improves class consistency, and enables more accurate condition assessment and distress-specific analysis. **True Labelling:** The depression and upheaval are annotated as separate instances according to their respective visible boundaries, even when the two distresses are directly adjacent. **False Labelling:** The depression and upheaval are merged into a single annotation region, obscuring the distinction between the two deformation mechanisms and reducing annotation consistency.